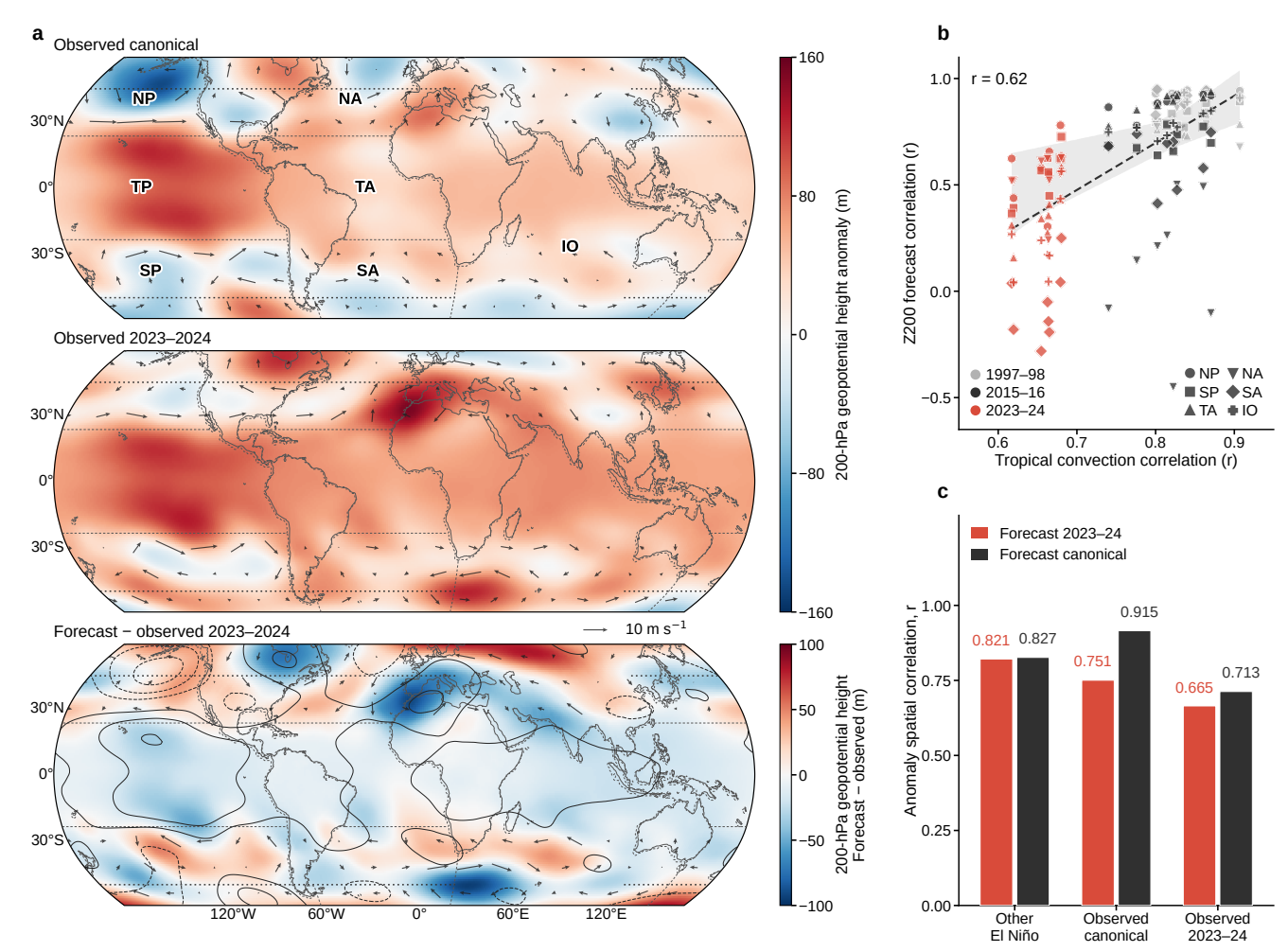

a
Observed canonical
NP
NA
TP
TA
IO
SP
SA
30°N
0°
30°S
Observed 2023–2024
Forecast − observed 2023–2024
10 m s−1
120°W
60°W
0°
60°E
120°E
200-hPa geopotential height anomaly (m)
160
80
0
−80
−160
200-hPa geopotential height Forecast − observed (m)
100
50
0
−50
−100
b
r = 0.62
Z200 forecast correlation (r)
Tropical convection correlation (r)
1997–98
2015–16
2023–24
NP
NA
SP
SA
TA
IO
c
Forecast 2023–24
Forecast canonical
Anomaly spatial correlation, r
0.821
0.827
0.751
0.915
0.665
0.713
Other El Niño
Observed canonical
Observed 2023–24